UTD AI/ML BOOTCAMP CAPSTONE

Predictive Maintenance for Small Machine Shops

A working edge platform that watches a machine, learns what normal looks like, and warns the owner before it fails.

Healthy Watch Fault

Sandra Hershey-Zagrans | Demo Day, November 12, 2026 | Live demo over Zoom

THE PROBLEM

Small shops run machines until they break, because the fix is priced for big plants

I visited small machine shops as research for this project and found the same pattern everywhere: good businesses, aging machines, and not one number about the health of the equipment.

An unplanned failure on a key machine means a scrapped part, a missed delivery, and an emergency repair bill, and in a small shop there is often no second machine.

Predictive maintenance exists. It is sold to factories with reliability engineers on staff. A five-person shop never gets the sales call.

COMMERCIAL SYSTEM

$15k to $40k

per deployment, plus subscription

THIS PROJECT

About $70

in sensors per machine, one mini PC per shop, no subscription

HOW IT IS BUILT

Sensor data in, a color out, and everything in between stays in the shop

1

Sensors

Vibration (MPU-6050 accelerometer), temperature, current clamp. Mounted on the machine.

2

ESP32 node

Computes features on the chip (RMS, peak) and posts compact readings over Wi-Fi.

3

Mini PC

FastAPI scores each reading with both models and stores it in PostgreSQL.

4

Dashboard

React, auto-refreshing. Every machine is green, yellow, or red.

Stack

Python, scikit-learn, FastAPI, PostgreSQL in Docker, React, Ubuntu, Arduino C++ on the ESP32

Why the edge

One mini PC on the shop floor runs all of it. No cloud, no subscription, and it keeps working when the internet does not.

THE DATA

Ten thousand machine readings, and only 3.4 percent of them are failures

The AI4I 2020 Predictive Maintenance dataset (Matzka, 2020), published for explainable machine learning in maintenance. Each row is one reading from a milling machine.

Input feature	What it tells us
Air and process temp	How hot the machine and the room are running
Rotational speed	Spindle RPM
Torque	How hard the motor is working
Tool wear	Minutes on the current tool
Engineered: temp delta, power	Two derived features the model found most useful

10,000
labeled readings

339
actual failures, about 3.4 percent

96.6%
accuracy of a model that never predicts a failure. Useless, and why accuracy is the wrong score.

TWO MODELS

One model knows the failures we have seen, the other notices the ones we have not

SUPERVISED

Gradient-boosted classifier

- Trained on the labeled AI4I readings
- Class weighting so the 3.4 percent is not ignored
- Decision threshold tuned for recall on the failure class
- Outputs a failure probability, mapped to green, yellow, or red
- Fast to score, which matters on a small edge box

UNSUPERVISED

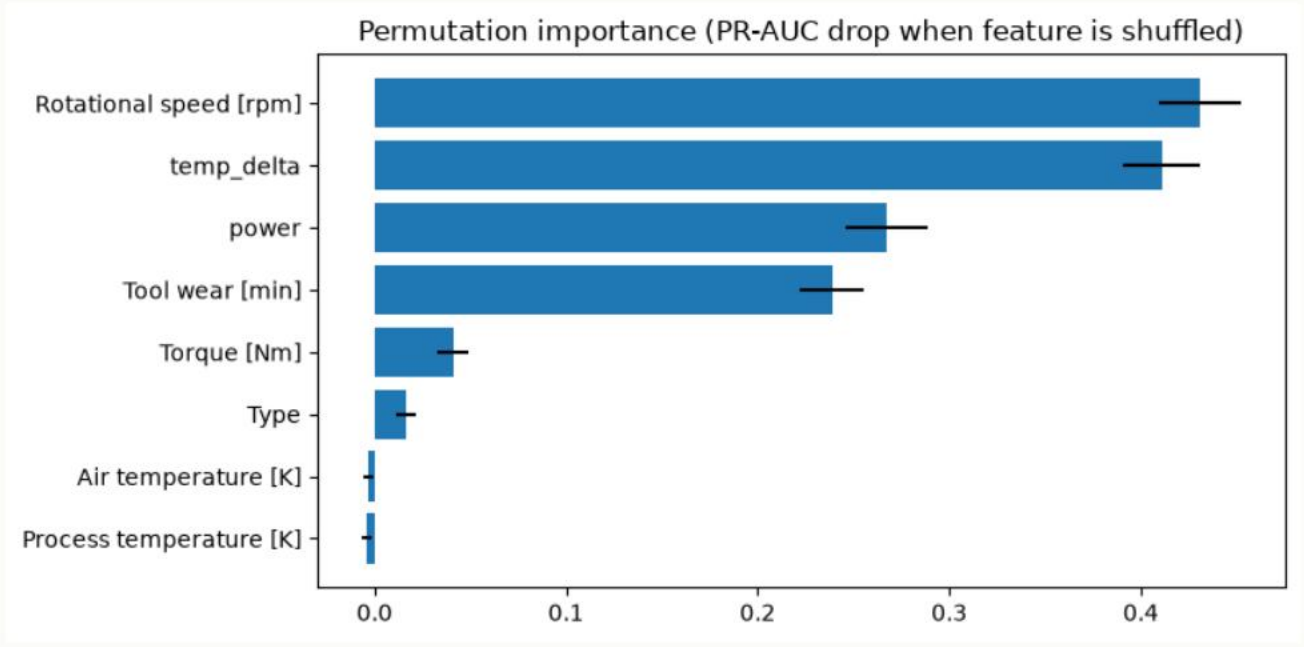
Isolation Forest anomaly detector

- Trained only on healthy readings, learns what normal looks like
- Flags anything that drifts far from that baseline
- Needs no failure labels at all
- The live rig applies the same idea by hand: a calibrated vibration baseline with watch and fault levels
- Catches the failure type nobody thought to label

Judged on precision, recall, and F1 for the failure class, plus the confusion matrix. Never on accuracy.

WHAT THE MODEL IS LOOKING AT

The model watches the same physics a good machinist feels



Speed and heat gap lead

Heat dissipation failures happen when the process-to-air gap is small at low speed. The model found that on its own.

Power and tool wear next

Torque times speed, and minutes on the tool. Both are things a machinist already watches.

Raw temperatures near zero

The engineered gap carries the signal, not the absolute readings. Explainable, not a black box.

RESULTS ON HELD-OUT DATA

What the classifier catches, and what it costs in false alarms

RECALL (FAILURES)

90.6%

77 of 85 real failures caught

PRECISION (FAILURES)

49.7%

About half of all alerts were real failures

F1 (FAILURES)

0.64

Balance of the two

THRESHOLD

0.046

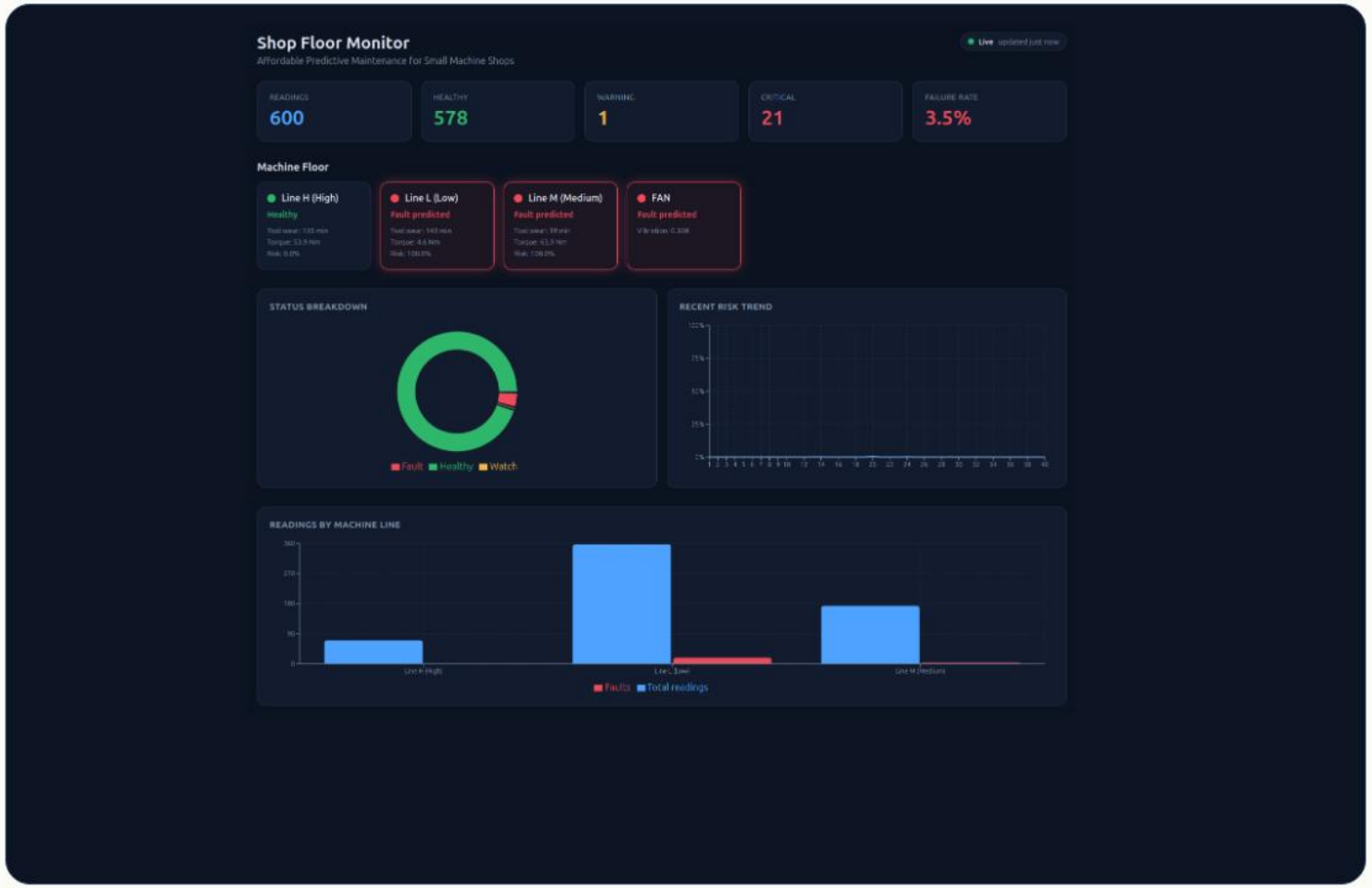
Set low on purpose, to favor catching failures over quiet dashboards

Confusion matrix	Predicted healthy	Predicted failure
Actually healthy (2,415)	2,337	78 false alarms
Actually failed (85)	8 missed	77 caught

The trade-off is deliberate. On 2,500 held-out readings, 78 false alarms is about one a day for a shop running steadily, each a five-minute look. The 8 misses are the cost that matters, so the threshold leans toward recall and the anomaly detector adds a second opinion on top.

THE WORKING SYSTEM

The owner does not read data. The owner sees a tile turn red and goes to look.



KPI cards
Machines online, alerts today, readings scored

Machine floor
One tile per machine, green, yellow, or red

Charts
Status breakdown, risk trend over time, readings by line

Live pulse
Auto-refresh every few seconds, with a last-updated indicator

LIVE DEMO

A real sensor, a real motor, and a fault I am about to cause on purpose

Healthy

The fan runs clean. The ESP32 streams a vibration reading every few seconds. It sits in the healthy band measured at calibration and the tile stays green.

Watch

I press a foam block into the blade path so the blades rub against it. Vibration climbs past the watch threshold and the tile goes yellow. This is the moment an owner schedules a look between jobs.

Fault

I push the foam in further. Vibration crosses the fault threshold, the tile turns red, and the chart shows the spike. Then I pull the foam out and watch it recover.

Everything you are watching runs on the mini PC on my desk. The only thing the internet is doing right now is carrying this Zoom call.

LIMITS AND NEXT STEPS

What it does not do yet, and what comes next

Three honest limits

- Point-in-time warning, not a countdown to failure
- Dataset features are not the rig features, so the rig is scored against a hand-calibrated vibration baseline
- A fan is a fair stand-in, but it is not a CNC

After the demo

- Bring the temperature and current sensors back in
- Fit the Isolation Forest on the rig's own vibration history, then on a real CNC
- Collect time-series history and work toward lead time

CLOSING

Small shops have been priced out of knowing their own machines. They do not need to be.

A $70 sensor, a mini PC, two models, and a dashboard an owner understands at a glance. Built, deployed, and running live today.

CODE

github.com/Alessandrarye/ai4i-capstone

DATASET

Matzka (2020), AI4I 2020 Predictive Maintenance, UCI Repository

Sandra Hershey-Zagrans | UTD AI/ML Bootcamp | Questions welcome